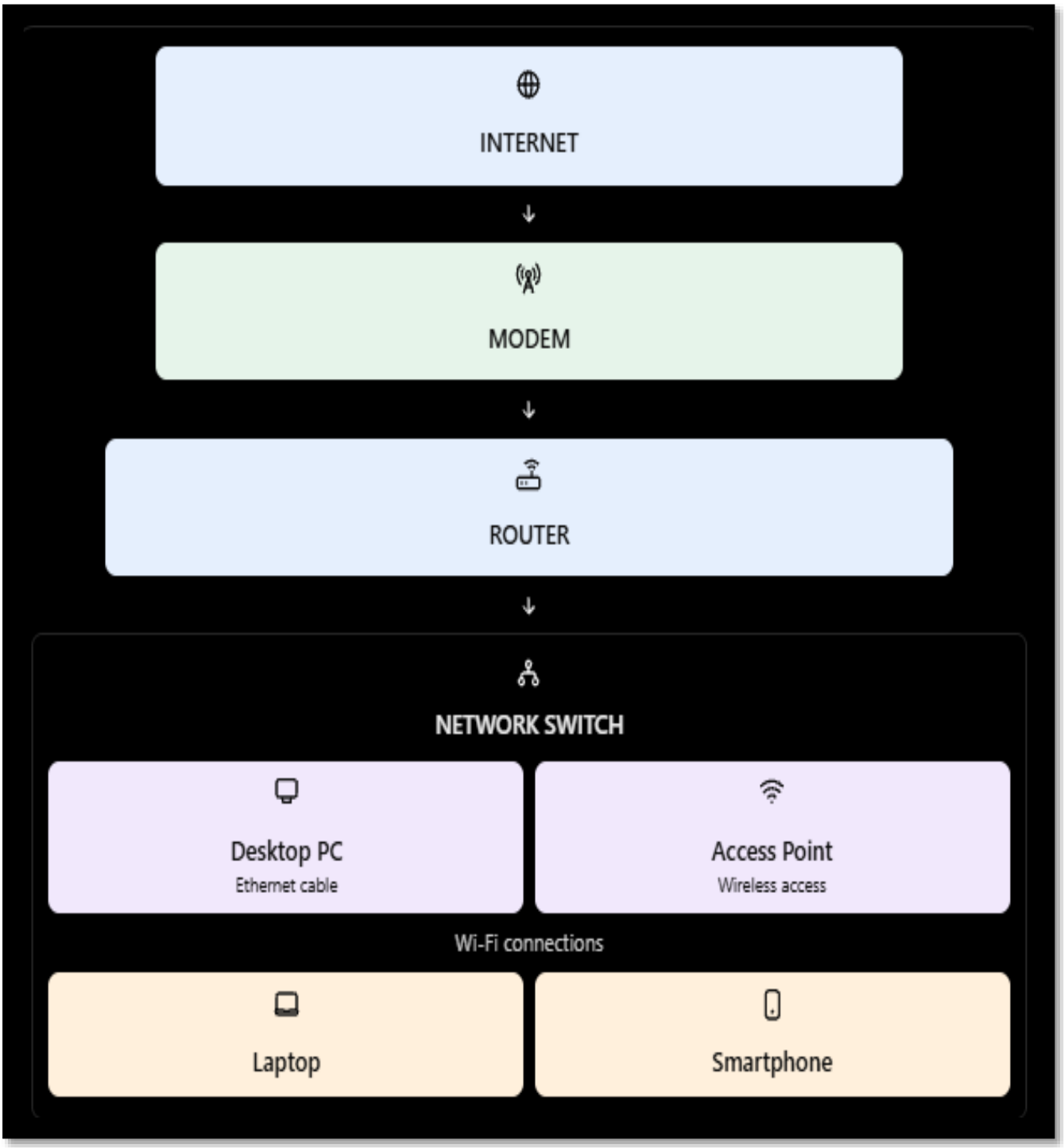
1. **Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.**



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it.

Device	Main function	Real-life example
Modem	Connects the home network to the internet service provider by converting signals as required by the connection technology.	A fiber ONT or cable modem supplied by an internet provider.
Router	Routes data between different networks and shares an internet connection among devices.	A Wi-Fi router used at home.
Switch	Connects multiple wired devices on the same local area network (LAN) and forwards data to the appropriate port.	A switch connecting computers in a college computer lab.
Access Point (AP)	Allows wireless devices to join a wired network through Wi-Fi.	An access point installed in an office or a large house.
Network Cable	Carries data signals between network devices through a physical connection.	An Ethernet cable connecting a desktop computer to a router.

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.



A. Ethernet Cable (Twisted Pair)

- Uses copper wires to transfer data between devices on a local network.
- Common use: Connecting computers, printers and routers in homes, offices and college labs.



B. Coaxial Cable

- Contains a central conductor surrounded by insulation and a protective metallic shield.
- Common use: Cable television and broadband internet connections.



C. Fiber Optic Cable

- Transmits data as pulses of light through thin glass or plastic fibers.
- Common use: High-speed broadband internet, telecommunications and long-distance network connections.

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Basis	Router	Switch
Main use	Connects different networks.	Connects devices within a local network.
Typical location	Homes, offices and internet gateways.	Computer labs, offices and server rooms.
Data handling	Forwards packets between networks using IP addresses and routing information.	Usually forwards Ethernet frames using MAC addresses.
Network layer	Primarily Layer 3 (Network Layer).	A typical Layer 2 switch operates at the Data Link Layer.
Internet connection	Commonly connects a local network to the internet.	Usually expands the number of wired ports in a local network.
Wireless capability	Many home routers include built-in Wi-Fi.	Standard switches generally do not provide Wi-Fi.
Ports	Often has a WAN port and several LAN ports.	Usually has multiple Ethernet ports for wired devices.
Example	A home Wi-Fi router sharing internet with phones and laptops.	An office switch connecting several desktop computers.